

How Graph Neural Networks Are Learning to Reverse-Engineer Small Brains' Wiring

A perspective on the research program of Allier, Heinrich, Lappalainen, Schneider & Saalfeld

Janelia Research Campus (HHMI) · Allen Institute · University of Tübingen

March 2026

Abstract

A research program spanning three years and four biological organisms has demonstrated that graph neural networks can reverse-engineer neural circuit connectivity from activity recordings alone. Starting from synthetic particle systems and advancing through simulated neural assemblies to biologically realistic connectomes of the fly visual system, the work reveals a counterintuitive principle: noise *helps* solve the inverse problem. The approach recovers not just who connects to whom, but synaptic strengths, time constants, neuron types, activation functions, and hidden stimuli—achieving $R^2 > 0.99$ on circuits with 13,741 neurons and 434,122 synapses. A mathematically rigorous degeneracy analysis shows precisely when and why the problem is solvable, while an LLM-driven experimental loop automates the scientific discovery process itself.

1 The Inverse Problem of Small Brains

Neuroscience has a data problem—but not the kind you might expect. Thanks to advances in electron microscopy and connectomics, we can now map the complete wiring diagram of entire nervous systems: every neuron, every synapse, every connection. The fruit fly *Drosophila melanogaster* had its full brain connectome published in 2024—roughly 140,000 neurons and 50 million synaptic connections, painstakingly traced from nanometer-resolution imagery.

But knowing *who* connects to *whom* is only half the story. A wiring diagram is like having the street map of a city without knowing the traffic patterns. To understand how a neural circuit actually computes—how it transforms sensory inputs into behavior—you need to know the *effective synaptic weights* (how strongly each connection operates), the *time constants* (how quickly each neuron responds), and the *nonlin-*

ear activation functions (how signals are transformed at each synapse).

Can you run the problem backward? Given recordings of neural activity—the “traffic”—can you infer the quantitative circuit model that generated it?

This is the question that Cedric Allier and collaborators at Janelia Research Campus have been systematically attacking since 2024. Their answer, refined across two published papers and an ambitious ongoing project spanning four biological organisms, is: *yes, under precise mathematical conditions that we can characterize exactly.*

2 From Particles to Neurons

The story begins not with neural circuits but with particles. In a 2024 paper [1], the team demonstrated that **message-passing graph neural networks** (GNNs) could decompose the dynamics of heterogeneous interacting systems into interpretable components. The key insight was architectural: a GNN’s computation—each node aggregating weighted messages from its neighbors—mirrors how physical systems actually work. Particles feel forces from neighbors. Neurons integrate synaptic inputs.

The framework was tested on seven systems: attraction-repulsion particles, gravitational dynamics, Coulomb interactions, flocking “boids,” wave propagation, reaction-diffusion patterns, and a signaling network. In every case, the GNN simultaneously recovered the interaction rules, the heterogeneous node properties (masses, charges, neuron types), and even the governing equations—the last step accomplished by feeding learned functions into symbolic regression, which returned $F \propto m/d^2$ and $F \propto q_1 q_2 / d^2$ without being told Newton’s or Coulomb’s laws.

The signaling network experiment—998 nodes, 17,865 edges, two node types—was the bridge. It

showed that a GNN trained to predict one-step derivatives of node activity could recover the full connectivity matrix ($R^2 = 1.00$) and classify node types with perfect accuracy.

3 Scaling to the Fly’s Small Brain

The 2026 follow-up [2] took this framework into neuroscience proper. The team designed a teacher-student paradigm: a biologically realistic ODE simulator (the “teacher”) generates neural activity from known circuit parameters, and a GNN (the “student”) learns to predict that activity while its learned edge weights are compared against ground truth.

The system studied is *not* a toy model. The *Drosophila* visual system simulation, built on the flyvis model of Lappalainen et al. (2024), comprises **13,741 neurons** across **65 cell types** with **434,122 synaptic connections**, organized into 217 retinotopic columns. Each neuron follows a non-spiking ODE:

$$\tau_i \frac{dv_i}{dt} = -v_i + V_{\text{rest},i} + \sum_j W_{ij} \cdot \text{ReLU}(v_j) + I_i(t) + \sigma \eta_i \quad (1)$$

where τ_i is the time constant, $V_{\text{rest},i}$ the resting potential, W_{ij} the synaptic weight from neuron j to i , $I_i(t)$ the visual stimulus, and $\sigma \eta_i$ process noise.

The GNN mirrors this structure with learnable components:

$$\frac{d\hat{v}_i}{dt} = f_\theta \left(v_i, \mathbf{a}_i, \sum_j \hat{W}_{ij} \cdot g_\phi(v_j, \mathbf{a}_j)^2, I_i \right) \quad (2)$$

where $\hat{W}_{ij}$ are learnable edge weights, g_ϕ and f_θ are MLPs (the “transfer” and “update” functions), and $\mathbf{a}_i \in \mathbb{R}^2$ are learnable per-neuron embeddings that capture cell-type identity without supervision.

4 The Noise Paradox

The most striking finding is counterintuitive: **adding noise to the neural dynamics dramatically improves connectivity recovery**.

At zero noise, the GNN achieves $R_W^2 = 0.90$ for synaptic weights—impressive, but imperfect. At moderate noise ($\sigma = 0.05$), recovery jumps to $R_W^2 = 0.96$. At high noise ($\sigma = 0.5$), it reaches $R_W^2 = 0.997$ —near-perfect recovery of all 434,122 synaptic weights, along with time constants ($R_\tau^2 = 1.00$), resting potentials ($R_{V_{\text{rest}}}^2 = 0.96$), and cell-type clustering accuracy of 0.91 across 65 types.

Why does noise help? The answer lies in a rigorous degeneracy analysis. The fly’s visual system is organized into retinotopic columns: the same 65 cell types are repeated across 217 spatial positions. In the noise-free regime, same-type neurons in different columns have nearly identical activation patterns. This creates a massive null space—approximately **121,100 of the 434,112 edge weights** (28%) can be changed freely without affecting the observed dynamics *at all*. The inverse problem is not merely difficult; it is mathematically ill-posed.

Process noise breaks this degeneracy. Each neuron receives independent stochastic perturbations, causing same-type neurons to diverge in their moment-to-moment activity. This lifts the rank of the activity matrix, collapsing the null space and making individual synaptic weights identifiable.

5 Four Organisms, One Framework

The current phase of the research—the connectome-gnn project—extends the approach to four biologically grounded neural circuits:

Circuit	Neurons	Edges	Dynamics
Fly visual system	13,741	434K	ReLU, graded
Fly central complex	152	9,722	Softplus ring attractor
Larva motor circuit	230	4,222	Two-population, gain-mod.
Zebrafish oculomotor	609	10,665	Pure linear integrator

Each circuit presents distinct challenges. The central complex is a ring attractor with softplus nonlinearity and per-neuron gains. The larva has a two-population architecture (premotor and motor neurons). The zebrafish is perhaps the hardest: its purely linear dynamics mean that infinitely many connectivity matrices produce identical activity, making the inverse problem fundamentally degenerate without noise.

Results across all four systems confirm the noise-helps principle. The zebrafish is the most dramatic example: connectivity R^2 rises from 0.02 (noise-free, fully connected) to 0.99 (noise = 0.5).

6 What You Can and Cannot Recover

A Procrustes-aligned SVD analysis of the learned connectivity matrices reveals a telling asymmetry. Decomposing $W = U\Sigma V^T$, the *output connectivity modes* U (which neurons co-activate) are consistently well-recovered ($R^2 \approx 0.90$ for the central complex), while the *input selection modes* V (which neurons drive which) are much harder ($R^2 \approx 0.51$).

This matches a theoretical prediction by Mastrogioiuseppe & Ostojic (2018): in recurrent networks, the output patterns are directly constrained by observed dynamics, while input selection is only indirectly observable. The U – V gap widens precisely when overall recovery is harder, providing a diagnostic for identifiability.

The method is also remarkably robust to corrupted prior knowledge. When the adjacency matrix is contaminated with 100–400% extra false edges, the GNN learns to assign near-zero weights to spurious connections, maintaining $R_W^2 > 0.94$. Even with 20% of true edges missing, recovery remains viable ($R_W^2 = 0.80$).

7 The LLM Scientist

Perhaps the most unconventional aspect of this research is how experiments are designed. The team built `GNN_LLM.py`, an automated pipeline where an LLM (Claude) acts as an iterative experimental scientist:

1. The LLM reads previous experimental results and accumulated principles.
2. It proposes a new hyperparameter configuration, reasoning about which parameters to change and why.
3. The experiment runs (data generation, GNN training, evaluation).
4. Results are fed back; the LLM updates its working memory.

Over 35 exploration campaigns (128 iterations each) across all four biomodels, this agentic loop discovered model-specific configurations that significantly outperform hand-tuned defaults: +22% R_W^2 for noise-free flyvis, +38% at moderate noise. It identified that Dale’s law regularization is essential for the central complex but destructive for the larva, that

learning rate on W and L1 sparsity are the dominant levers, and that many architectural parameters (MLP depth, hidden dimension) have surprisingly flat dose-response curves.

The LLM doesn’t just tune hyperparameters—it accumulates *causal understanding* of why certain regimes succeed or fail, discovering phenomena like bimodal convergence landscapes and sharp augmentation cliffs that would be invisible to grid search.

8 Beyond Simulation: The Road Ahead

The ultimate goal is real neural data: calcium imaging, electrophysiology, light-sheet recordings. The team has already begun experiments with the Zap-bench zebrafish electrophysiology dataset and is preparing a NeurIPS 2025 submission targeting the full four-organism benchmark with baseline comparisons (MLP, RNN, Neural ODE, state-space models).

Critical open questions remain. Can the framework handle time-dependent connectivity (synaptic plasticity)? Time delays? The gap between voltage traces and calcium indicators? Missing neurons? The degeneracy analysis provides a precise mathematical framework for answering these questions: each additional biological complication either expands or contracts the null space, and the answer depends on the specific circuit architecture.

References

- [1] Allier, C., Schneider, M.C., Innerberger, M., Heinrich, L., Bogovic, J.A., & Saalfeld, S. (2024). Decomposing heterogeneous dynamical systems with graph neural networks. *arXiv:2407.19160*.
 - [2] Allier, C., Heinrich, L., Schneider, M., & Saalfeld, S. (2026). Graph neural networks uncover structure and functions underlying the activity of simulated neural assemblies. *arXiv:2602.13325*.
 - [3] Lappalainen, J.K., et al. (2024). Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634, 1132–1140.
 - [4] Beiran, M. & Litwin-Kumar, A. (2023). Parametric control of flexible timing through low-dimensional neural manifolds. *Neuron*, 111, 739–753.
-

The Hidden Dimensions of Neural Circuits

There is a structural analogy between the situation facing this research program and the one facing string theory. In physics, general relativity operates in four dimensions—three of space and one of time—and describes gravity with extraordinary precision at all accessible scales. String theory proposes that reality has ten dimensions: the familiar four, plus six extra spatial dimensions curled up at the Planck scale ($\sim 10^{-35}$ m), far too small to observe directly. The four-dimensional theory is the *effective* theory that emerges after these small dimensions are compactified: it captures the low-energy physics faithfully, but it cannot represent the degrees of freedom that were integrated out.

For three years, this research program has operated in a directly analogous regime. The state of each neuron is described by a single scalar: the membrane voltage $v_i(t) \in \mathbb{R}$. But a real neuron is not a single number. Its full dynamical state includes at least four distinct classes of hidden variables:

1. **Receptor-specific currents.** Each synapse carries multiple receptor types—AMPA ($\tau \sim 1$ ms), NMDA ($\tau \sim 50$ ms), GABA ($\tau \sim 10$ ms)—each with its own kinetics, reversal potential, and voltage dependence. The scalar ODE collapses all of these into a single summed input $\sum_j \mathbf{W}_{ij} \text{ReLU}(v_j)$.
2. **Temporal history.** Axonal conduction delays, synaptic facilitation, and short-term depression mean that the message neuron j sends to neuron i depends not on the current voltage $v_j(t)$ alone, but on the trajectory $\mathbf{v}_j(t) = [v_j(t), v_j(t-1), \dots]^\top$. The scalar ODE sees only $v_j(t)$.
3. **Adaptation and intrinsic excitability.** Calcium-activated potassium channels, spike-frequency adaptation, and after-hyperpolarization create neuron-internal memory that modulates the response to incoming drive. These are absent from the leak-integrator equation.
4. **Neuromodulatory state.** Dopamine, serotonin, acetylcholine, and norepinephrine reconfigure effective synaptic weights on timescales of seconds to minutes. A single real-valued $\mathbf{W}_{ij}$ cannot represent a synapse whose sign and strength depend on context.

These hidden variables are the “extra dimensions” of neural dynamics. They are real and measurable in principle—with patch clamp, optogenetics, or pharmacology—but *invisible to the instruments that generate large-scale datasets*. Calcium imaging, the dominant recording technology, reports a slow (~ 100 ms decay), nonlinear, non-negative proxy: a convolution of the true voltage with an indicator kernel, sampled at video frame rates. Neuropixels electrodes are faster but sparse. In both cases, the observation is a low-dimensional projection of a high-dimensional reality—exactly as the four-dimensional physics we measure is a projection of string theory’s ten.

The compactification is not metaphorical. When a calcium indicator integrates a burst of fast synaptic currents (~ 1 ms AMPA, ~ 50 ms NMDA, ~ 10 ms GABA) into a single slow fluorescence transient, it is performing precisely the operation that compactifies a small dimension: it rolls up the fast temporal structure into a single observable, erasing the information about *which* receptor types fired in *which* sequence. The scalar-voltage ODE used in this work— $\tau_i \dot{v}_i = -v_i + V_i^{\text{rest}} + \sum_j \mathbf{W}_{ij} \text{ReLU}(v_j) + I_i$ —is the effective field theory that emerges after this compactification. It captures the slow dynamics faithfully, but it cannot represent the fast degrees of freedom that were integrated out.

What would the “extra dimensions” of neural dynamics look like? Two candidates are already taking shape. The first is **temporal depth**: replacing the scalar voltage $v_j(t)$ with a history vector $\mathbf{v}_j(t) \in \mathbb{R}^T$ of past activity, and introducing learned temporal filters $h_\xi(\mathbf{a}_i, \mathbf{v}_j)$ that compress the sender’s history in a receiver-dependent way—generalizing axonal delay kernels into a structure that maps onto the transformer’s Query-Key-Value architecture, with the sparse connectome $\widehat{\mathbf{W}}_{ij}$ replacing the attention mask. The second is **complex-valued connectivity**: promoting $\mathbf{W}_{ij} \in \mathbb{R}$ to $W_{ij}^{\mathbb{C}} = |W_{ij}| e^{i\varphi_{ij}} \in \mathbb{C}$, so that a single neuromodulatory parameter α can reconfigure all N^2 synapses simultaneously via $W_{ij}^{\text{eff}}(\alpha) = |W_{ij}| \cos(\varphi_{ij} - \alpha)$. This rescues Feynman-style interference at the message-passing step—synaptic signals can constructively and destructively combine depending on their phase alignment—and yields falsifiable predictions: two behavioral contexts determine all others, and antipodal contexts produce sign-reversed connectivity.

Neither extension has been tested against real data. Like string theory's extra dimensions, they are motivated by mathematical structure and biological plausibility rather than direct observation. The hidden variables—axonal delays, receptor-specific time constants, neuromodulatory phases—are real, measurable in principle, but invisible to the recording modalities that generate the datasets on which the GNN is trained. The question that will determine the next phase of this research is the same question that haunts string theory: do the compactified dimensions leave signatures in the low-energy observables? If the fast synaptic structure imprints itself on the slow calcium dynamics—through timing-dependent motifs, stimulus-specific gain changes, or context-dependent connectivity patterns that a scalar ODE cannot explain—then the higher-dimensional theory makes predictions that the four-dimensional one does not. If it does not, the extra dimensions may be mathematically coherent but experimentally inert: beautiful, and invisible.